TEJAS SACHDEVA

tejas.sach@gmail.com | Portfolio | LinkedIn | (346) 218-9443

EDUCATION

The University of Texas at Austin – Austin, TX
Master of Science in Mechanical Engineering May 2027
Bachelor of Science in Mechanical Engineering | GPA: 3.8 May 2026

EXPERIENCE

Biotex, Inc. – Houston, TX May 2026 – Aug 2026
Mechanical Engineering Intern — Oncomagnetics (Non-Invasive Glioblastoma Therapy Wearable)

- Designed CNC-machined 6061-T6 assembly fixtures for motor subassemblies in a non-invasive glioblastoma therapy helmet; applied poka-yoke and DFMA principles to mechanically constrain components to a single orientation; sized locating features from worst-case tolerance analysis of the subassembly interfaces to hold press-fit alignment across 500+ units
- Authored ASME Y14.5 GD&T production drawings for motor subassembly components; defined inspection criteria tying critical features to assembly fit requirements
- Redesigned the helmet retention dial with a textured grip pattern; moved to MJF-printed PA11 after supplier DFM review indicated that knurl pressing induced material failure in machined plastic; validated snap-fit engagement with worst-case stack-ups
- Redesigned the head cradle to conform to a 95th-percentile head model; specified BASF TPU for skin-contact biocompatibility and MJF over vacuum casting after casting tooling was rated for only 15 cycles, cutting per-unit cost and lead time

General Motors – Warren, MI May 2025 – Aug 2025
Service Release Engineering Intern (Chassis, Body Exterior)

- Drove $700K+ in cost savings across 2023–2025 Equinox and Terrain platforms by consolidating duplicate service components into shared part numbers
- Analyzed part geometry and critical design features in Siemens Teamcenter to qualify components as interchangeable across vehicle platforms
- Partnered with product and supply chain teams to execute BOM simplification, reducing service lifecycle complexity

Parker Wellbore – Houston, TX May 2024 – Aug 2024
Technical Services Intern

- Designed a specialized crane basket in SolidWorks to safely transport BOP stacks on drilling rigs, producing 3D models and technical drawings for fabrication
- Performed FEA (linear static, modal, fatigue) per API, ASME, and AISC standards to determine FOS and WLL, selecting materials via Charpy V-notch testing to reduce Von-Mises stress fluctuation 65% under impact loading

ORGANIZATIONS

Texas Robotics – Austin, TX Dec 2025 – Present
Research Assistant

- Engineered load-bearing 3D-printed carbon fiber test fixtures aligning an upper-limb exoskeleton to a harmonic drive-actuated arm dummy at the elbow and wrist torque points, holding a 200 Nm/rad impedance boundary under dynamic loading
- Enabled accurate kinematic data capture from the test bench, characterizing the assistive torque required to support average human arm motion

Longhorn Racing Solar – Austin, TX Aug 2024 – Aug 2025
Powertrain Engineer

- Designed battery cooling manifolds in SolidWorks and ran ANSYS Fluent thermal simulations, achieving 25% improvement in temperature uniformity and 35% reduction in pressure drop under race conditions

LEADERSHIP AND COMMUNITY INVOLVEMENT

First-Year Experience, UT Austin – Austin, TX May 2024 – Jan 2025
First-Year Interest Group (FIG) Mentor

- Mentored a cohort of 20+ first-year students through academic and social transition to college, delivering individualized guidance on course planning and campus integration

ASME Purdue University Chapter – West Lafayette, IN Oct 2022 – May 2023
Treasurer

- Managed $20K+ in funds and secured external sponsorships to support engineering competitions and initiatives

ADDITIONAL

- **Technical Skills:** SolidWorks, Autodesk Fusion 360, AutoCAD Mechanical, GD&T, DFMA, Tolerance Analysis, Siemens Teamcenter, ANSYS (Fluent, Mechanical), MATLAB, Python, LabVIEW
- **Honors & Certifications:** Richard Douglas and Judith Watson Perkins Endowed Presidential Scholar; Design for Six Sigma (Green Belt)
- **Patents:** Multi-Spectral Retinal Imaging & Virtual Spectral Decomposition, U.S. Patent App. No. 19/553,050